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(54) **SYSTEMS AND METHODS FOR CONTROLLING AN INVERTER OF A TOWED ELECTRIC VEHICLE**

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Primary Examiner — Tuan C To

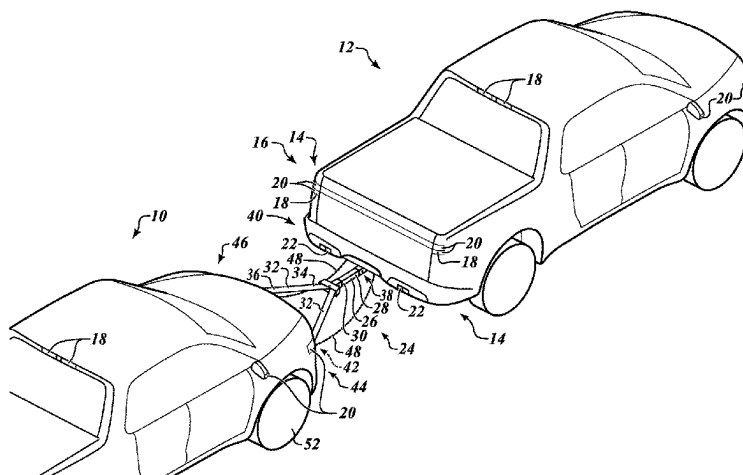
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(57) **ABSTRACT**

Various disclosed embodiments include systems, vehicles, and methods for controlling an inverter of a towed vehicle. In an illustrative embodiment, a system includes a controller. The controller includes a processor and computer-readable media configured to store computer-executable instructions configured to cause the processor to: receive sensed data indicative of detected deceleration of a tow vehicle; and

(Continued)



during detected deceleration, control an inverter of a towed vehicle responsive to the detected deceleration.

20 Claims, 14 Drawing Sheets

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B60T 17/22 (2006.01)
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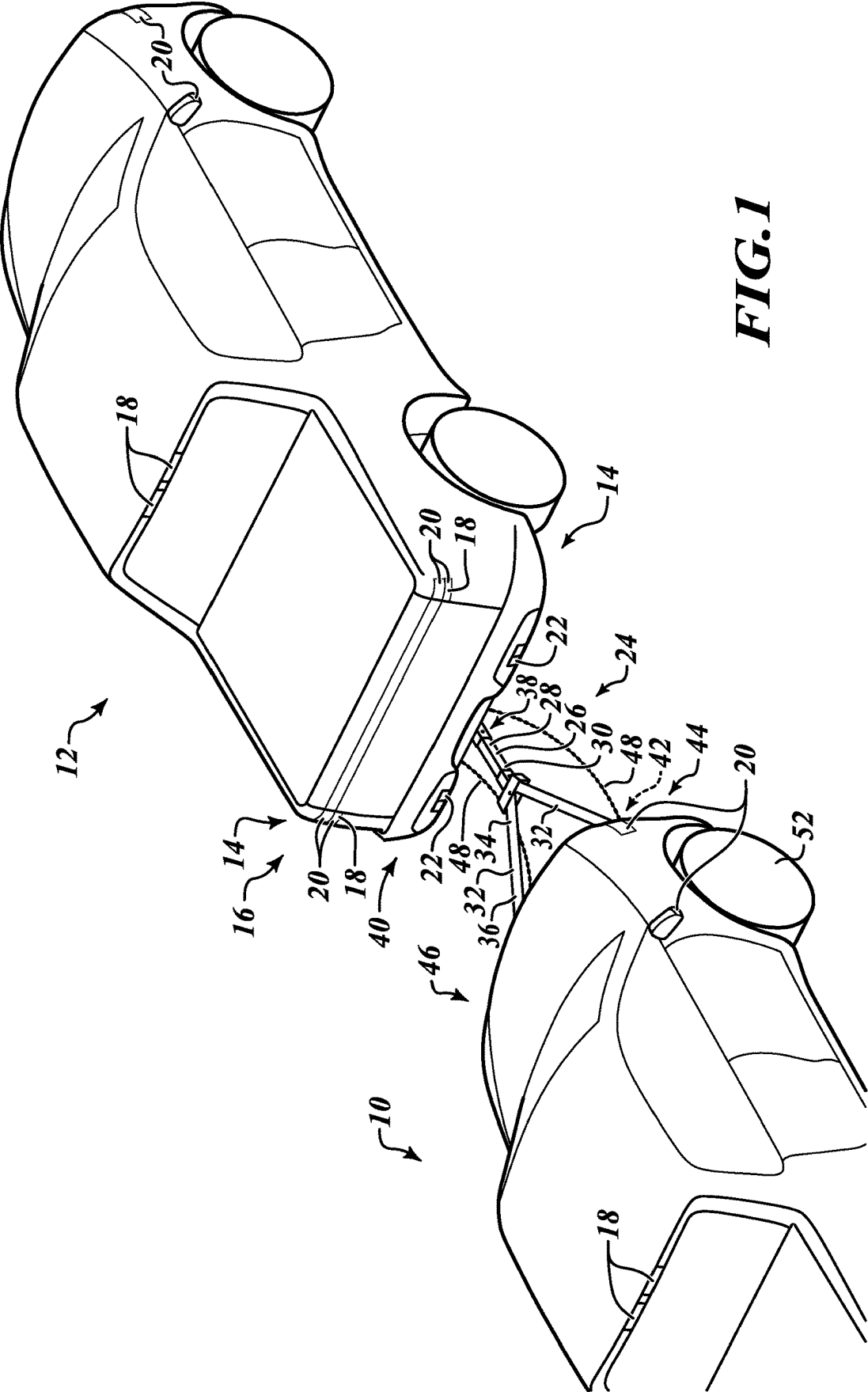


FIG. 1

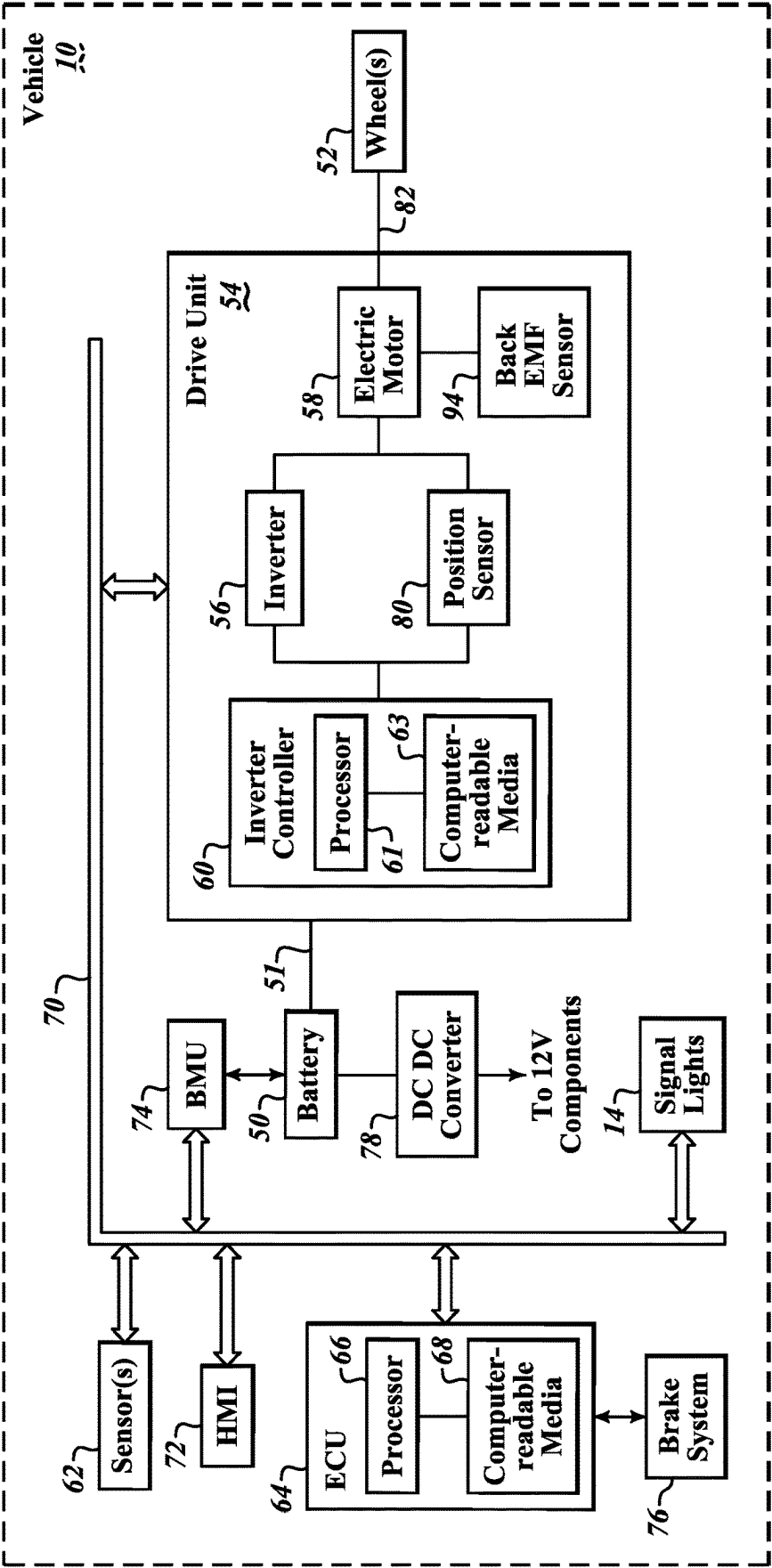


FIG. 2A

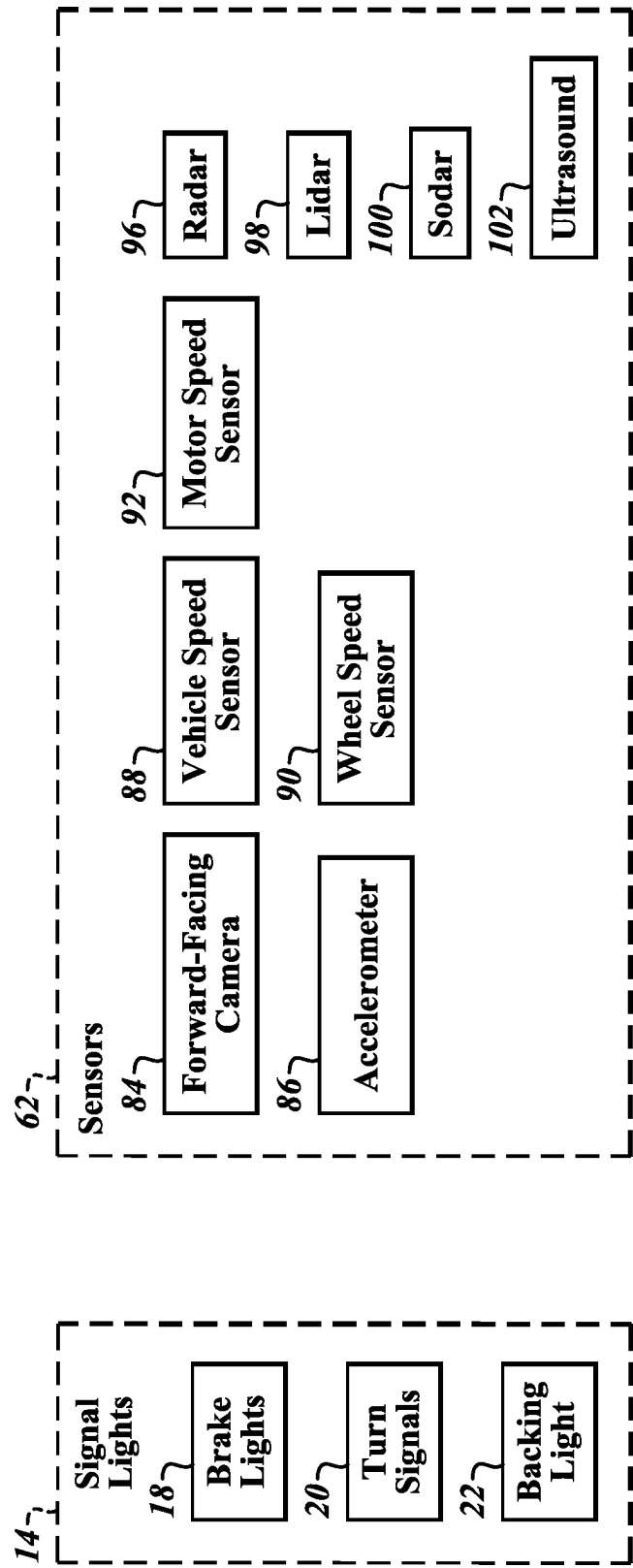


FIG.2C

FIG.2B

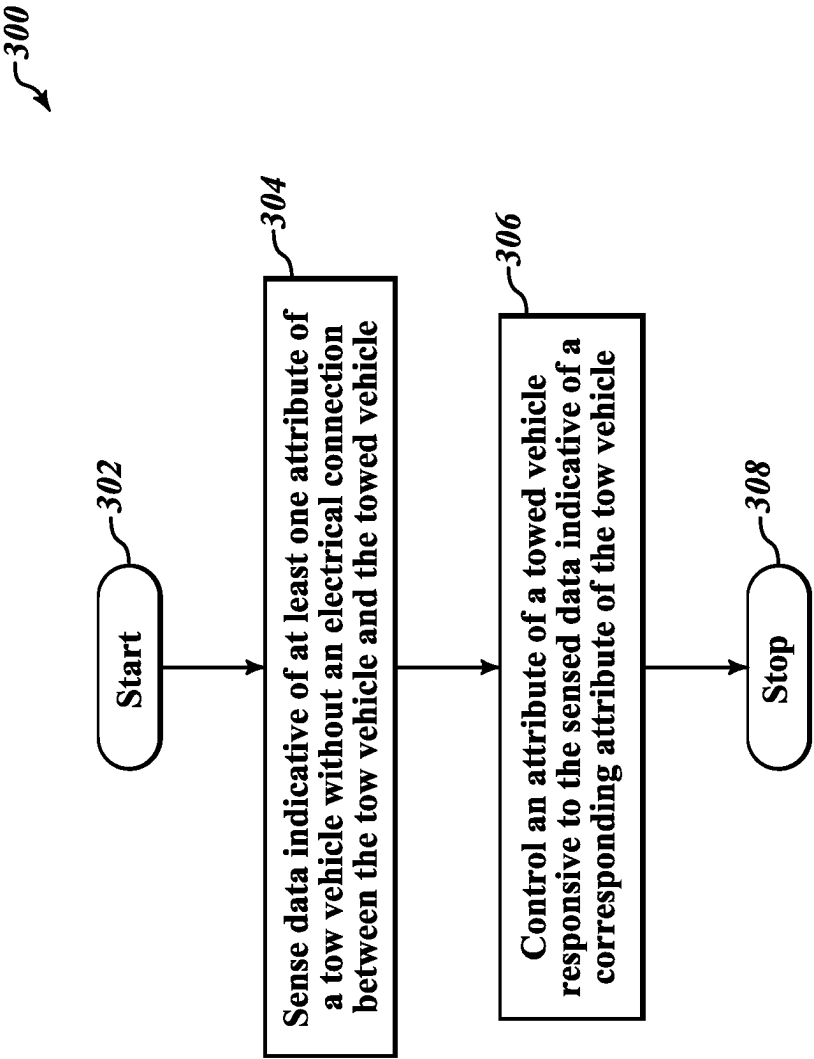
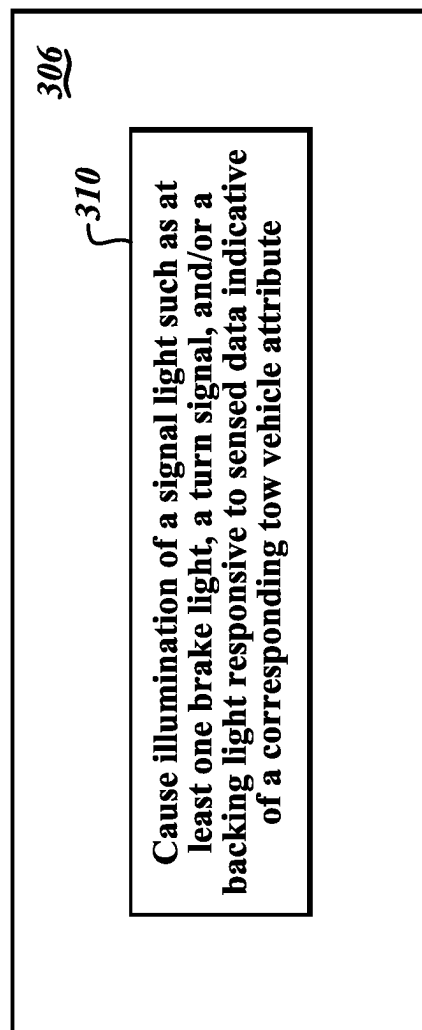
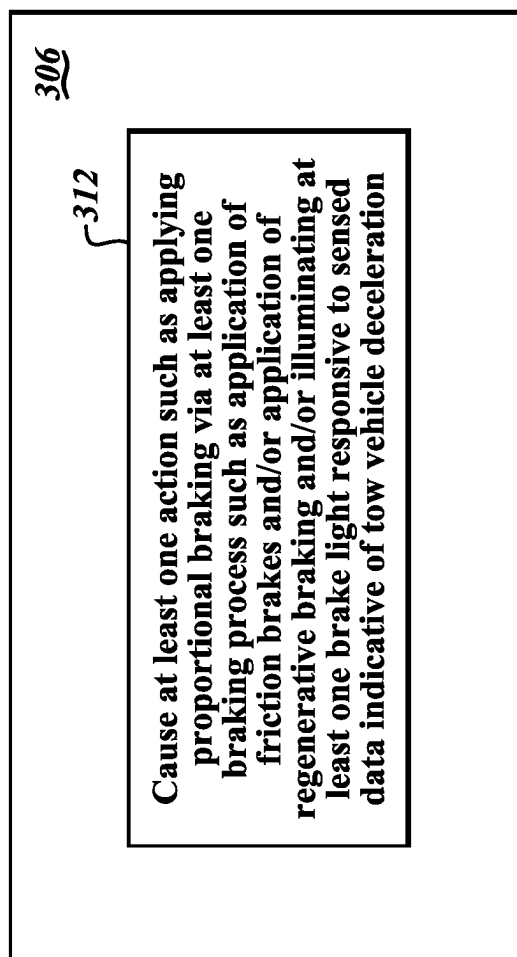
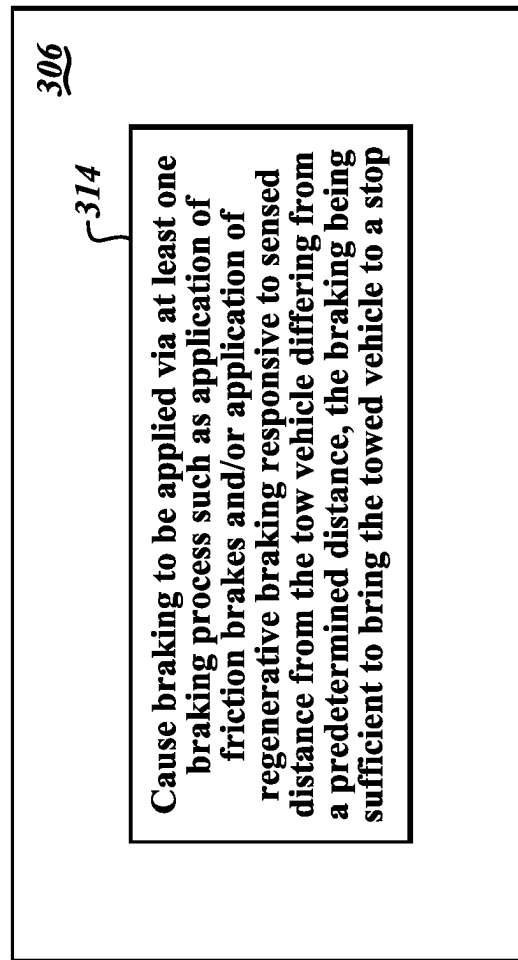


FIG.3A

**FIG.3B**

***FIG. 3C***

***FIG. 3D***

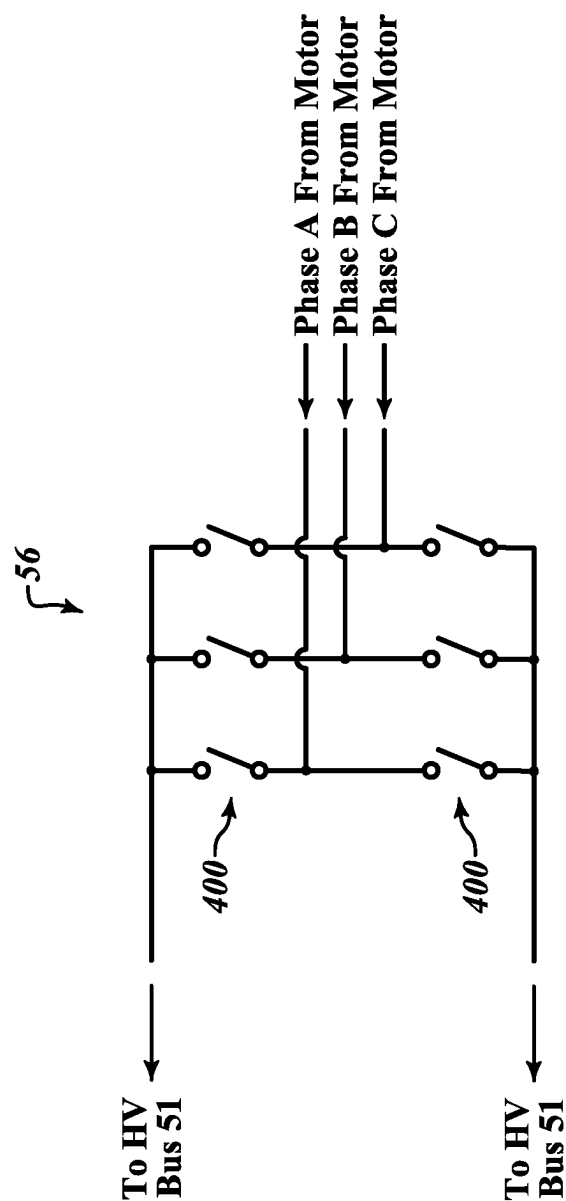


FIG.4

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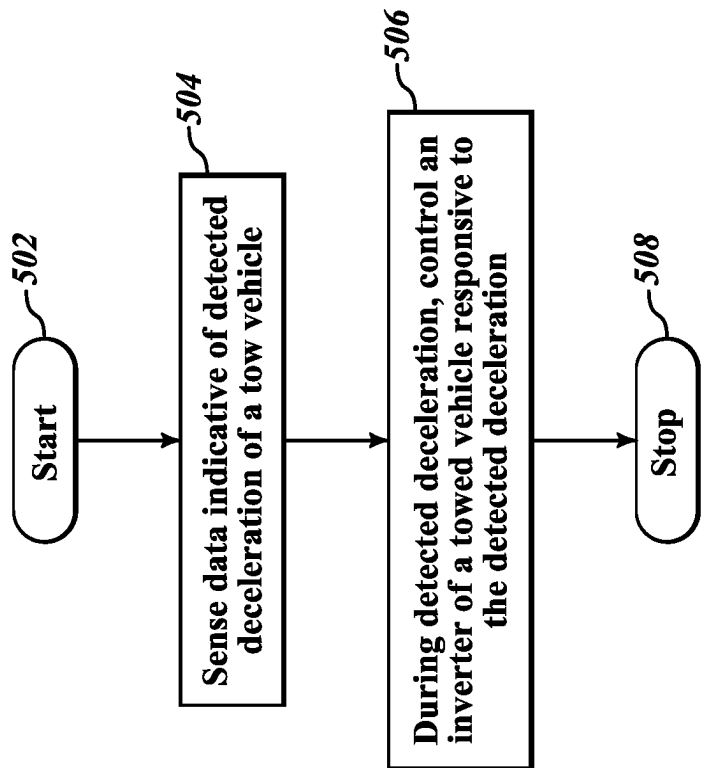
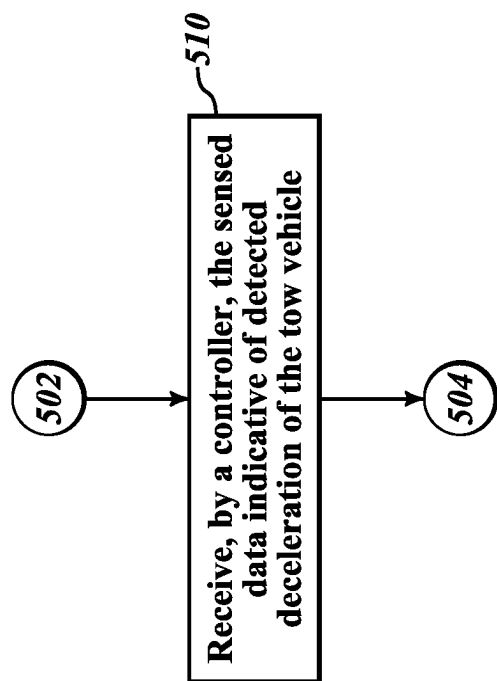
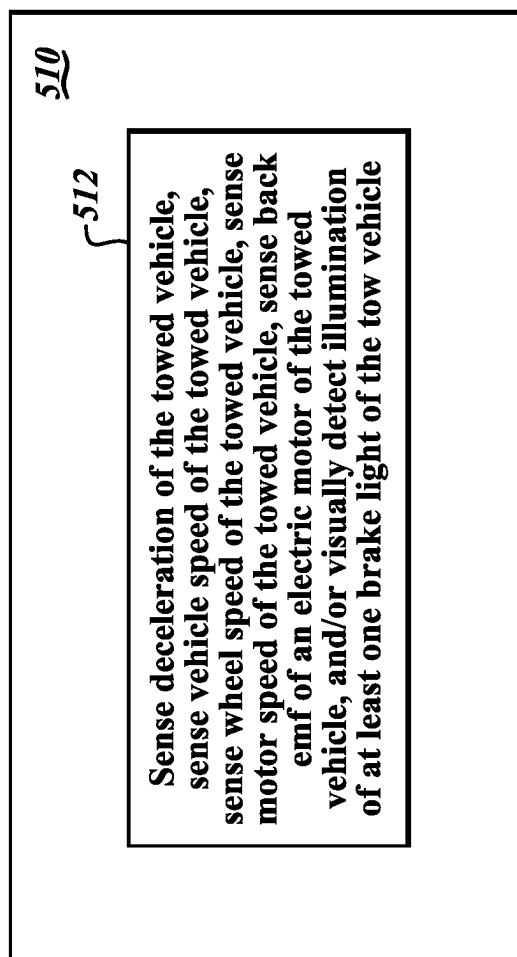
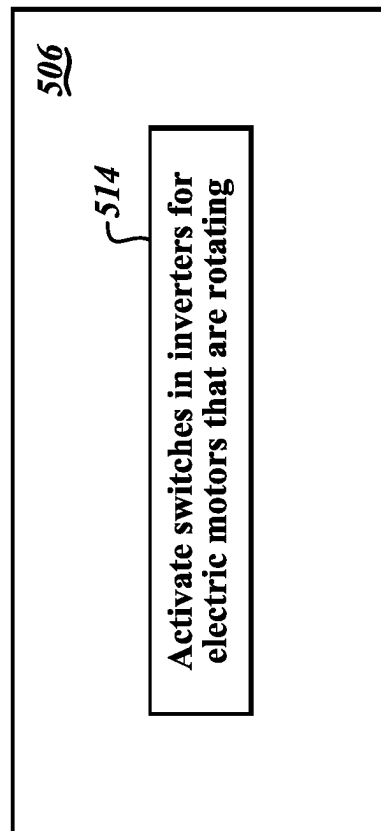
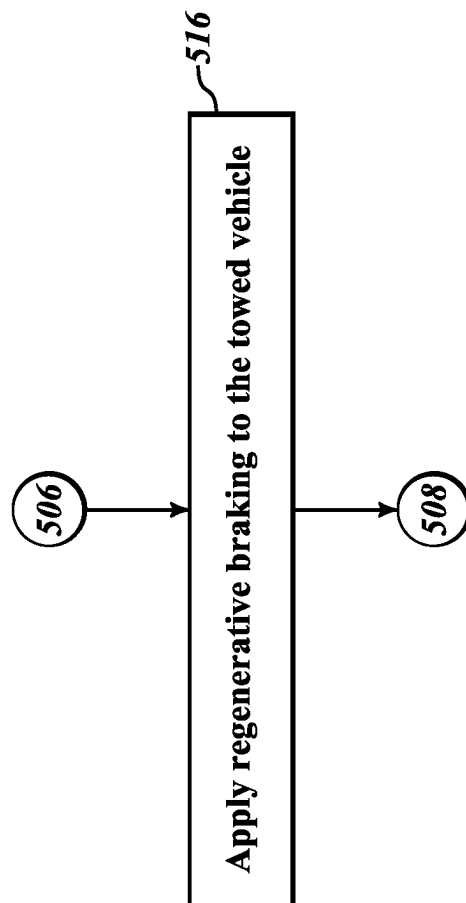


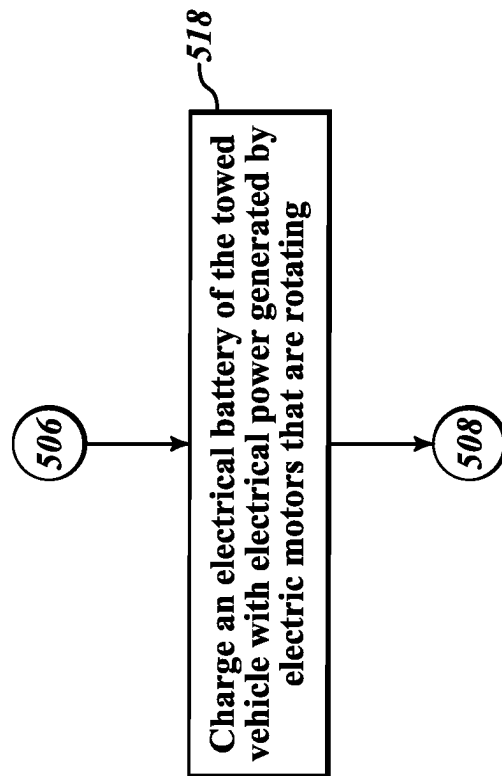
FIG. 5A

**FIG. 5B**

***FIG. 5C***

***FIG. 5D***

***FIG. 5E***

***FIG. 5F***

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SYSTEMS AND METHODS FOR CONTROLLING AN INVERTER OF A TOWED ELECTRIC VEHICLE

CROSS-REFERENCE TO RELATED APPLICATION

The present disclosure claims the benefit of priority of co-pending International Application No. PCT/US21/50854, filed on Sep. 17, 2021, and entitled "SYSTEMS AND METHODS FOR CONTROLLING AN INVERTER OF A TOWED ELECTRIC VEHICLE," which claims priority to U.S. Provisional Patent Application No. 63/079,846, filed on Sep. 17, 2020, and entitled "SYSTEMS AND METHODS FOR TOWING AN ELECTRIC VEHICLE WITHOUT EXTERNAL ELECTRICAL CONNECTION TO A TOW VEHICLE, SELECTIVE REGENERATIVE BRAKING DURING DECELERATION IN A TOWED ELECTRIC VEHICLE, AND POWERING LOW VOLTAGE ELECTRICAL LOADS IN A TOWED ELECTRIC VEHICLE," the contents of which are incorporated in full by reference herein.

INTRODUCTION

The present disclosure relates to towing an electric vehicle.

When a motor vehicle is being towed behind a tow vehicle, signal lights (that is, brake lights, turn signals, and backing lights) and braking of the towed vehicle currently are synchronized with the tow vehicle via an external electrical connection between the tow vehicle and the towed vehicle and with aftermarket equipment, such as auxiliary braking systems.

When an electric vehicle driven by permanent magnet synchronous motors is towed, back electromotive force (back emf) is generated that, beyond a certain motor speed, may be higher than voltage of a high voltage high voltage (HV) direct current (DC) bus and, if uncontrolled, could cause energy to flow back into the battery in an uncontrolled manner. Further, uncontrolled back emf that is not consumed into the battery may subject inverter components or other components on the HV DC bus to voltages that may be greater than their designed voltage ranges.

Moreover, constant, uncontrolled application of regenerative braking torque in a towed electric vehicle can cause additional fuel to be consumed by the tow vehicle and can possibly add to instability of the towed electric vehicle by adding sway.

The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

BRIEF SUMMARY

Various disclosed embodiments include systems, vehicles, and methods for controlling an inverter of a towed vehicle.

In an illustrative embodiment, a system includes a controller. The controller includes a processor and computer-readable media configured to store computer-executable instructions configured to cause the processor to: receive sensed data indicative of detected deceleration of a tow vehicle; and during detected deceleration, control an inverter of a towed vehicle responsive to the detected deceleration.

In another illustrative embodiment, a vehicle includes rotatable wheels and at least one drive unit including: at least

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one inverter; and at least one electric motor electrically couplable to the at least one inverter and rotatably couplable to at least one of the rotatable wheels. The vehicle also includes at least one sensor and a controller. The controller includes a processor and computer-readable media configured to store computer-executable instructions configured to cause the processor to: receive from the at least one sensor sensed data indicative of detected deceleration of a tow vehicle; and during detected deceleration, control the at least one inverter responsive to the detected deceleration.

In another illustrative embodiment, a method includes: sensing data indicative of detected deceleration of a tow vehicle; and during detected deceleration, controlling an inverter of a towed vehicle responsive to the detected deceleration.

The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

Illustrative embodiments are illustrated in referenced figures of the drawings. It is intended that the embodiments and figures disclosed herein are to be considered illustrative rather than restrictive.

FIG. 1 is a perspective view of an illustrative electric vehicle being towed by a tow vehicle.

FIG. 2A is a block diagram of the illustrative electric vehicle of FIG. 1.

FIG. 2B is a block diagram of illustrative signal lights of the vehicle of FIGS. 1 and 2.

FIG. 2C is a block diagram of illustrative sensors of the vehicle of FIGS. 1 and 2A.

FIG. 3A is a flow chart of an illustrative method of controlling towing of a vehicle without an electrical connection between a tow vehicle and a towed vehicle.

FIGS. 3B-3D are flow charts of details of the method of FIG. 3A.

FIG. 4 is a block diagram of an illustrative inverter.

FIG. 5A is a flow chart of an illustrative method of controlling an inverter of a towed vehicle.

FIGS. 5B-5F are flow charts of details of the method of FIG. 5A.

Like reference symbols in the various drawings generally indicate like elements.

DETAILED DESCRIPTION

In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here.

Various disclosed embodiments include systems, vehicles, and methods for controlling an inverter of a towed vehicle.

By way of nonlimiting overview and referring to FIGS. 1 and 2A, in various embodiments an illustrative system includes a controller 64. The controller 64 includes a processor 66 and computer-readable media 68 configured to

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store computer-executable instructions configured to cause the processor 66 to receive sensed data indicative of detected deceleration of a tow vehicle 12 and, during detected deceleration, control an inverter 56 of a towed vehicle 10 responsive to the detected deceleration.

Illustrative details regarding the vehicle 10 (to be towed) will be explained first by way of illustration only and not of limitation. Then, illustrative details regarding controlling the towing of the vehicle 10 without an electrical connection between the tow vehicle 12 and the towed vehicle 10 and controlling the inverter 56 of the towed vehicle 10 will be explained by way of illustration only and not of limitation.

The present disclosure refers to “deceleration.” As used herein, deceleration is negative acceleration.

As shown in FIG. 1, the vehicle 10 is towable behind 12 tow vehicle 12. In various embodiments the vehicle 10 may be any suitable electric vehicle, such as, for example, an electric vehicle, a hybrid vehicle, a plug-in hybrid vehicle, and the like. However, it will be appreciated that, as explained below, in some embodiments the vehicle 10 may be any type motor vehicle whatsoever regardless of source of propulsion and may also include vehicles powered by an internal combustion engine, such as a spark-ignition engine or a compression-ignition engine, a fuel cell, natural gas, propane, or the like. Regardless of propulsion source, the vehicle 10 may be any type of towable motor vehicle whatsoever as desired, such as, for example and without limitation, a car, a truck, a sport-utility vehicle, a van, a motorhome, a bus, or the like.

In various embodiments, the tow vehicle 12 may be any type of motor vehicle (that is capable of towing another motor vehicle) whatsoever as desired, such as, for example and without limitation, a car, a truck, a sport-utility vehicle, a van, a motorhome, a bus, or the like. The tow vehicle 12 includes various signal lights 14 disposed about an aft end 16 of the tow vehicle 12. The signal lights 14 are configured to illuminate and convey information to others regarding actions of the tow vehicle 12 and intentions of a driver of the tow vehicle 12. For example, the signal lights 14 suitably include brake lights 18, turn signals 20, and a backing light 22.

The vehicle 10 is suitably tethered to the tow vehicle 12 via a tow bar 24. The tow bar 24 includes a shank 26 having a forward end 28 and an aft end 30. The tow bar 24 also includes arms 32 that each have a forward end 34 and an aft end 36. The arms 32 are attached at their forward ends 34 to the aft end 30 of the shank 26.

A hitch receiver 38 is attached to a lower portion 40 of the aft end 16 of the tow vehicle 12. A base plate (not shown for purposes of clarity) having a cross member 42 is attached to a lower portion 44 of a front end 46 of the vehicle 10. The hitch receiver 38 evenly distributes towing loads to a frame (not shown for purposes of clarity) of the tow vehicle 12 and the base plate evenly distributes towing loads to a frame (not shown for purposes of clarity) of the vehicle 10. The forward end 28 of the shank 26 is removably attached to the hitch receiver 38 and the aft ends 36 of the arms 32 are removably attached to the base plate. Safety cables 48 (such as steel cables or chains) suitably are removably attached to the hitch receiver 38 and the base plate to help prevent accidental and/or inadvertent disconnection of the towed vehicle 10 from the tow vehicle 12.

As will be explained below, in various embodiments towing of the vehicle 10 can be controlled without an electrical connection between the tow vehicle 12 and the vehicle 10 and, in various embodiments, the inverter 56 of the vehicle 10 can likewise be controlled without an elec-

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trical connection between the tow vehicle 12 the vehicle 10. As such, it will be appreciated that, in various embodiments, the vehicle 10 can be towed behind the tow vehicle 12 at highway speeds without an electrical connection between the tow vehicle 12 the vehicle 10 and without use of aftermarket equipment.

As shown in FIG. 2A, in various embodiments and as an initial, high-level introductory example the vehicle 10 includes a high-voltage, direct current (DC) electrical battery 50, rotatable wheels 52, and at least one drive unit 54. Each drive unit 54 suitably includes at least one inverter 56 electrically couplable to the battery 50. Each inverter 56 suitably includes three-terminal power semiconductor devices (not shown in FIG. 2A). Each drive unit 54 also suitably includes at least one electric motor 58 electrically couplable to the at least one inverter 56 and rotatably couplable to at least one of the rotatable wheels 52. Each drive unit 54 also includes at least one inverter controller 60 configured to control the at least one inverter 56. The vehicle 10 also includes at least one sensor 62 and a controller 64 (such as an electronics control unit or ECU). The controller 64 includes a processor 66 and computer-readable media 68 configured to store computer-executable instructions.

Now that an initial, high-level introductory example of the vehicle 10 has been set forth, additional details will be explained by way of non-limiting examples provided by way of illustration only and not of limitation.

As also shown in FIG. 2A, in various embodiments the battery 50 includes a high-voltage DC electrical battery. In such embodiments, the battery 50 is configured to provide high-voltage DC electrical power, such as on the order of around 450 volts or so via a high voltage (HV) DC bus 51. In various embodiments the battery 50 may include a lithium-ion battery. However, it will be appreciated that the battery 50 may include any suitable battery as desired and that further description of the battery is not necessary for a person of skill in the art to understand disclosed subject matter.

In various embodiments the vehicle 10 includes the electronics control unit (ECU) 64 that controls operations of various components via a peer-to-peer network bus 70 such as a controller area network (CAN) bus. Other peer-to-peer network buses, such as a local area network (LAN), a wide area network (WAN), or a value-added network (VAN), may also be used for enabling communication between the ECU 64 and the components connected to the peer-to-peer network bus 70.

In various embodiments the processor 66 may include a computer processing unit (CPU), a general purpose processor, a digital signal processor, a field programmable gate array, or the like, and/or any combination thereof. Processors are well known and further description of their construction and operation are not necessary for an understanding by a person of skill in the art of disclosed subject matter.

In various embodiments the computer-readable media 68 may include any suitable computer memory configured to store computer-executable instructions configured to cause the processor 66 to perform functions described herein. Given by way of non-limiting examples, the computer-readable media 68 may include any suitable volatile memory elements, such as without limitation random access memory (RAM), such as dynamic RAM (DRAM), static RAM (SRAM), static-dynamic RAM (SDRAM), and the like, nonvolatile memory elements such as without limitation read-only-memory (ROM), hard drive, tape, compact-disc ROM (CDROM), and the like, and combinations thereof.

Moreover, the computer-readable media **18** may incorporate electronic, magnetic, optical, and/or other types of storage media as desired.

In various embodiments the ECU **64** communicates via the peer-to-peer network bus **70** with a human-machine interface (HMI) **72**. In various embodiments and given by way of example only and not of limitation, the HMI **72** may include mechanical buttons or switches or may include selectable graphical user interface features presented on a vehicle display device(s).

In various embodiments the ECU **64** communicates via the peer-to-peer network bus **70** with a battery management unit (BMU) **74**. In various embodiments and given by way of example only and not of limitation, the BMU **74** communicates with the battery **50** to generate battery status information, which is sent to the ECU **64** via the peer-to-peer network bus **70**. The BMU **74** receives battery information from the battery **50** and/or from any sensors (not shown for purposes of clarity) associated with or included in the battery **50**. The battery information may include state of charge (SOC), temperature, voltage of battery cells, input/output current, coolant flow, or other values related to battery operations. The BMU **74** uses the battery information to control battery recharging and battery thermal management and to communicate with other components of the vehicle **10** via the peer-to-peer network bus **70** or with external components or other systems or devices as desired.

In various embodiments, the ECU **64** communicates with a brake system **76** directly or via the peer-to-peer network bus **70**. In various embodiments the brake system **76** may include a foot pedal (not shown for purposes of clarity), a brake solenoid or pressure sensor (not shown for purposes of clarity), a handbrake (not shown for purposes of clarity), a steering wheel-mounted brake paddle (not shown for purposes of clarity), or comparable brake components (not shown for purposes of clarity), and friction brakes (not shown for purposes of clarity). In various embodiments, if desired, the brake system **76** may be a brake-by-wire system or any type of brake system as desired.

In various embodiments, a DC/DC converter **78** is electrically coupled to the battery **50** to provide DC electrical power at an appropriate DC voltage level (such as, for example and without limitation, around 12 VDC or so) to various auxiliary loads that are to be energized while the vehicle **10** is being towed, such as various control units, external lighting systems, coolant pumps for cooling the drive units **54** and their components, and the like.

In various embodiments the drive unit **54** may include one or more inverters **56**, one or more position sensors **80** such as a resolver, and one or more electric motors **58**, such as without limitation brushless direct current (BLDC) motors, alternating current induction motors (ACIM), permanent magnet (PM) synchronous motors (PMSM), interior PM motors (IPMM), PM switch reluctance motors (PMSRM), or any suitable electric motors whatsoever as desired. A back emf sensor **94** is configured to sense and measure back emf of an associated electric motor **58** when the electric motor **58** is functioning as a generator.

The drive unit **54** also includes at least one inverter controller **60** electrically coupled to the inverter(s) **56** and the position sensor(s) **80**. The inverter controller **60** includes a processor **61** and computer-readable media **63** configured to store computer-executable instructions configured to cause the processor **61** to perform functions described herein. In various embodiments, the ECU **64** communicates with the inverter controller **60** via the peer-to-peer network bus **70** to cause the inverter controller **60** to perform various

functions described herein. A drive member **82**, such as an axle or the like, is rotatably coupled to the electric motor **58**. At least one propulsion wheel **52** is rotatably couplable to the drive member **82**.

It will be appreciated that, if desired, one or more of the wheels **52** suitably may be rotatably disconnectable as desired from an associated electric motor **58**, thereby preventing the associated motor **58** from rotating while the vehicle **10** is being towed. However, in various embodiments at least one electric motor **58** is rotatable by its associated wheel **52** while the vehicle **10** is towed, thereby providing for application of regenerative braking and charging of the battery **50** as desired and as further explained below.

Referring additionally to FIG. 2B, in various embodiments the signal lights **14** of the vehicle **10** may also include, without limitation, at least one of the brake lights **18**, at least one of the turn signals **20** (FIG. 1), and the backing light **22**. In various embodiments, the ECU **64** communicates with the signal lights **14** via the peer-to-peer network bus **70** to perform various functions described herein.

Referring additionally to FIG. 2C, in various embodiments the sensors **62** may include any sensor desired for a particular application. For example and by way of non-limiting example, in various examples the sensors **62** may include, without limitation, a forward-facing camera **84**, an accelerometer **86** (that senses fore/aft acceleration and/or lateral acceleration), a vehicle speed sensor **88**, a wheel speed sensor **90**, and a motor speed sensor **92**. Also, in various embodiments the sensors **62** may include, without limitation, a radio detection and ranging (RADAR) sensor **96**, a light detection and ranging (LIDAR) sensor **98**, a sound detection and ranging (SODAR) sensor **100**, an ultrasound sensor **102**, and the like. In various embodiments, the ECU **64** communicates with the sensors **62** via the peer-to-peer network bus **70** to perform various functions described herein.

Now that the vehicle **10** has been described, controlling towing of the vehicle **10** without an electrical connection between the tow vehicle **12** and the vehicle **10** will be explained by way of non-limiting examples given by way of illustration only and not of limitation.

In various embodiments the computer-executable instructions are configured to cause the processor **66** to receive from the at least one sensor **62** sensed data indicative of at least one attribute of the tow vehicle **12** without an electrical connection between the tow vehicle **12** and the vehicle **10** and to cause the processor **66** to control an attribute of the vehicle **10** responsive to the sensed data indicative of a corresponding attribute of the tow vehicle **12**.

In some such embodiments, the signal lights **14** of the tow vehicle **12** and the signal lights **14** of the vehicle **10** can be synchronized. It will be appreciated that in such embodiments the vehicle **10** may be any type motor vehicle whatsoever regardless of source of propulsion and may also include vehicles powered by an internal combustion engine, such as a spark-ignition engine or a compression-ignition engine, a fuel cell, natural gas, propane, or the like (in addition to the electric vehicles described above).

In such embodiments, the tow vehicle attribute includes visual signaling, the sensor **62** includes a forward-facing camera **84**, and the data indicative of the tow vehicle attribute includes at least one datum such as visually detected illumination of at least one brake light **18** of the tow vehicle **12**, visually detected illumination of a turn signal **20** of the tow vehicle **12**, and visually detected illumination of a backing light of the tow vehicle **12**. It will be appreciated

that, in various embodiments, data from the camera **84** may include image data and/or video data.

In such embodiments, the instructions are further configured to cause the processor **66** to cause illumination of a signal light of the vehicle **10**, such as at least one brake light **18**, a turn signal **20**, and a backing light **22** responsive to data indicative of a corresponding attribute (that is, illumination of a corresponding signal light **14**) of the tow vehicle **12**.

In some other embodiments, braking of the tow vehicle **12** and braking of the vehicle **10** can be synchronized. It will be appreciated that in some such embodiments (discussed below) the vehicle **10** may be any type motor vehicle whatsoever regardless of source of propulsion and may also include vehicles powered by an internal combustion engine, such as a spark-ignition engine or a compression-ignition engine, a fuel cell, natural gas, propane, or the like (in addition to the electric vehicles described above).

In some such embodiments, the tow vehicle attribute includes deceleration and the at least one sensor **62** may include any one or more of the accelerometer **86**, the vehicle speed sensor **88**, the wheel speed sensor **90**, the motor speed sensor **92**, the back emf sensor **94**, and/or the forward-facing camera **84**.

In such embodiments, the data indicative of the tow vehicle attribute includes at least one datum such as sensed deceleration of the vehicle **10** from the accelerometer **86**, sensed vehicle speed of the vehicle **10** from the vehicle speed sensor **88**, sensed wheel speed of the vehicle **10** from the wheel speed sensor **90**, sensed motor speed of the vehicle **10** from motor speed sensor **92**, sensed back emf of one or more of the electric motors **58** of the vehicle **10** from the back emf sensor **94**, and visually detected illumination of at least one brake light **18** of the tow vehicle **12** from the forward-facing camera **84**.

In various such embodiments the instructions are further configured to cause the processor **66** to cause at least one action such as applying proportional braking via at least one braking process like application of friction brakes, application of regenerative braking, and illuminating at least one brake light responsive to data indicative of tow vehicle deceleration.

Illustrative details regarding synchronization of the brake light **18** of the tow vehicle **12** and the brake light of the vehicle **10** have been discussed above and are not repeated for sake of brevity.

In some such embodiments, responsive to data indicative of tow vehicle deceleration the processor **66** causes the brake system **76** to apply proportional braking via application of friction brakes. In such embodiments, the friction brakes may be part of a brake-by-wire system or part of any other type of brake system as desired.

It will be appreciated that, in embodiments that synchronize the brake light **18** of the tow vehicle **12** and the brake light of the vehicle **10** and that synchronize proportional braking via application of friction brakes, the vehicle **10** may be any type motor vehicle whatsoever regardless of source of propulsion and may also include vehicles powered by an internal combustion engine, such as a spark-ignition engine or a compression-ignition engine, a fuel cell, natural gas, propane, or the like (in addition to the electric vehicles described above).

In some other such embodiments, responsive to data indicative of tow vehicle deceleration the processor **66** causes the brake system **76** to apply proportional braking via application of regenerative braking. In such embodiments the ECU **64** communicates with the inverter controller **60** via the peer-to-peer network bus **70** to cause the processor

61 to control the inverter **56** to apply proportional regenerative braking. It will be appreciated that application of regenerative braking can help contribute to reducing wear of friction brakes of the vehicle **10**, thereby helping contribute to increasing life of friction brakes of the vehicle **10**. Details regarding regenerative braking will be discussed further below by way of illustration only and not of limitation.

In some other embodiments the vehicle **10** may be brought to a safe stop if the physical tow connection between the tow vehicle **12** and the vehicle **10** is broken. That is, the vehicle **10** may be brought to a safe stop if, when the vehicle **10** is being towed behind the tow vehicle **12**, the tow bar **24** breaks, or if the shank **26** of the tow bar **24** becomes disconnected from the hitch receiver **38** of the tow vehicle **12**, or if one or more of the arms **32** of the tow bar **24** become disconnected from the cross member **42**.

In such embodiments the tow vehicle attribute includes distance from the tow vehicle **12** to the vehicle **10** and the at least one sensor **62** includes at least one distance sensor such as the radar sensor **96**, the LIDAR sensor **98**, the SODAR sensor **100**, and the ultrasound sensor **102**. It will be appreciated that such sensors may be included in an Advanced Driver Assistance Systems (ADAS) suite of sensors. It will also be appreciated that some such sensors may also be referred to as front parking sensors and/or adaptive cruise control sensors. In such embodiments the data indicative of the tow vehicle attribute includes at least one datum such as sensed distance from the tow vehicle.

In such embodiments the instructions are further configured to cause the processor **66** to cause braking to be applied via at least one braking process such as application of friction brakes and/or application of regenerative braking responsive to sensed distance from the tow vehicle **12** differing from a predetermined distance, and the braking is sufficient to bring the vehicle **10** to a stop.

For example, in some such embodiments the predetermined distance may be the distance between the tow vehicle **12** and the vehicle **10** when the vehicle **10** is tethered to the tow vehicle with the tow bar **24**. If, when the vehicle **10** is being towed behind the tow vehicle **12**, the tow bar **24** breaks, or if the shank **26** of the tow bar **24** becomes disconnected from the hitch receiver **38** of the tow vehicle **12**, or if one or more of the arms **32** of the tow bar **24** become disconnected from the cross member **42**, then the distance between the tow vehicle **12** and the vehicle **10** will change.

For example, in some instances the towed vehicle **12** and the vehicle **10** will get farther away from each other as the vehicle **10** coasts and slows but the tow vehicle **12** continues to be driven. As another example, in some other instances the towed vehicle **12** and the vehicle **10** will get closer each other as the vehicle **10** coasts and slows (decelerates) but the tow vehicle **12** brakes or otherwise slows (decelerates) faster than the rate of slowing (or deceleration) of the vehicle **10**. Regardless, in either case, the processor **66** causes braking to be applied via at least one braking process such as application of friction brakes and application of regenerative braking responsive to sensed distance from the tow vehicle **12** differing from the predetermined distance, and the braking is sufficient to bring the vehicle **10** to a stop—thereby helping to secure a “runaway” vehicle **10**.

Referring additionally to FIG. 3A, in various embodiments a method **300** is provided for controlling towing of a towed vehicle without an electrical connection between a tow vehicle and the towed vehicle. The method **300** starts at a block **302**. At a block **304** data indicative of at least one attribute of a tow vehicle is sensed without an electrical connection between the tow vehicle and the towed vehicle.

At a block **306** an attribute of a towed vehicle is controlled responsive to the sensed data indicative of a corresponding attribute of the tow vehicle. The method **300** stops at a block **308**.

Referring additionally to FIG. 3B, in various embodiments the tow vehicle attribute includes visual signaling and the sensed data indicative of the tow vehicle attribute includes at least one datum such as visually detected illumination of at least one brake light, visually detected illumination of a turn signal, and/or visually detected illumination of a backing light. In such embodiments, controlling an attribute of a towed vehicle responsive to the sensed data indicative of a corresponding attribute of the tow vehicle at the block **306** includes causing illumination of a signal light such as at least one brake light, a turn signal, and/or a backing light responsive to sensed data indicative of a corresponding tow vehicle attribute at a block **310**.

Referring additionally to FIG. 3C, in various embodiments the tow vehicle attribute includes deceleration and the sensed data indicative of the tow vehicle attribute includes at least one datum such as sensed deceleration of the towed vehicle, sensed vehicle speed of the towed vehicle, sensed wheel speed of the towed vehicle, sensed motor speed of the towed vehicle, sensed back emf of an electric motor of the towed vehicle, and/or visually detected illumination of at least one brake light of the tow vehicle. In such embodiments, controlling an attribute of a towed vehicle responsive to the sensed data indicative of a corresponding attribute of the tow vehicle at the block **306** includes causing at least one action such as applying proportional braking via at least one braking process such as application of friction brakes and/or application of regenerative braking and/or illuminating at least one brake light responsive to data indicative of tow vehicle deceleration at a block **312**.

Referring additionally to FIG. 3D, in various embodiments the tow vehicle attribute includes distance between the tow vehicle and the towed vehicle and the sensed data indicative of the tow vehicle attribute includes at least one datum chosen from sensed distance from the tow vehicle. In such embodiments, controlling an attribute of a towed vehicle responsive to the sensed data indicative of a corresponding attribute of the tow vehicle at the block **306** includes causing braking to be applied via at least one braking process such as application of friction brakes and/or application of regenerative braking responsive to sensed distance from the tow vehicle differing from a predetermined distance, the braking being sufficient to bring the towed vehicle to a stop at a block **314**.

Now that various embodiments of controlling towing of the vehicle **10** without an electrical connection between the tow vehicle **12** and the vehicle **10** have been explained by way of illustration only and not of limitation, various embodiments of controlling the inverter **56** of the vehicle **10** will be set forth below by way of illustration only and not of limitation.

In various embodiments the computer-executable instructions are configured to cause the processor **66** to receive sensed data indicative of detected deceleration of the tow vehicle **12** and, during detected deceleration, control the inverter **56** of the vehicle **10** responsive to the detected deceleration—such as to apply regenerative braking during the detected deceleration and/or to charge the battery **50** during the detected deceleration.

In some such embodiments, the at least one sensor **62** includes at least one sensor such as the accelerometer **86**, the vehicle speed sensor **88**, the wheel speed sensor **90**, the motor speed sensor **92**, the back emf sensor **94**, and the

camera **84**. In such embodiments, the sensed data indicative of detected deceleration of the tow vehicle can include data such as sensed deceleration of the vehicle **10** from the accelerometer **86**, sensed vehicle speed of the vehicle **10** from the vehicle speed sensor **88**, sensed wheel speed of the vehicle **10** from the wheel speed sensor **90**, sensed motor speed of the vehicle **10** from motor speed sensor **92**, sensed back emf of one or more of the electric motors **58** of the vehicle **10** from the back emf sensor **94**, and visually detected illumination of at least one brake light **18** of the tow vehicle **12** from the forward-facing camera **84**.

Referring additionally to FIG. 4, in various embodiments the inverter **56** includes switches **400**. In various embodiments the switches **400** may include any suitable inverter switch as desired, such as without limitation metal-oxide-semiconductor (MOS) field-effect transistors (FETs) (MOSFETs) and/or insulated-gate bipolar transistors (IGBTs), or the like, as desired for a particular application. The switches **400** are shown for purposes of illustration only and are not intended to represent any particular switching condition or configuration. It will be appreciated that the switches **400** may be switched via drive signals provided to gate terminals (not shown for purposes of clarity) of the switches **400** from a gate drive circuit (not shown for purposes of clarity) within or driven by the inverter controller **60**. In various embodiments, the gate terminals of the switches **400** may be driven via any modulation scheme as desired, such as, without limitation, pulse width modulation or the like.

Given by way of illustration only and not of limitation, during towing of the towed vehicle **10** when deceleration is not detected, the switches **400** are activated by the inverter controller **60** to generate zero torque and to minimize back emf with respect to speed, thereby helping allow the vehicle **10** to be towed at higher speeds (such as at highway speeds). In various embodiments, a small amount of electrical energy is put into the electric motor **58** to cancel losses generated by the electric motor **58** being rotated in order to effect zero mechanical output from the electric motor **58**. As such, in such embodiments the battery **50** is electrically connected to the HV DC bus **51** while the vehicle **10** is being towed, thereby helping prevent components of the inverter **56** and components of the HV DC bus **51** from being subjected to voltages that may be greater than their designed voltage ranges.

Also given by way of illustration only and not of limitation, in various embodiments, during deceleration, deceleration is detected as described above. In response to the detected deceleration, the ECU **64** issues a negative torque request (proportional to the amount of detected deceleration) to the inverter controller **60**. Energy is produced by the electric motor **58** when subjected to the negative torque request. This imparts negative torque at the drive member **82** (that is, regenerative braking), which produces electrical energy that is returned to the battery **50**, thereby charging the battery **50**.

In view of the illustrative examples set forth above, in various embodiments the computer-executable instructions are configured to cause the processor **66** to, during detected deceleration, control the inverter **56** of the vehicle **10** responsive to the detected deceleration—such as to apply regenerative braking during the detected deceleration and/or to charge the battery **50** during the detected deceleration. Thus, in various embodiments the battery **50** is not charged when deceleration is not detected and regenerative braking is not applied when deceleration is not detected.

In various embodiments the instructions are further configured to cause the processor **66** to, responsive to the

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detected deceleration and during the detected deceleration, cause the inverter controller **60** to control activation of the switches **400**, during the detected deceleration, in inverters **56** for electric motors **58** that are rotatable during towing. In some such embodiments, the instructions are further configured to cause the processor **66** to, responsive to the detected deceleration, apply regenerative braking during the detected deceleration. As described above, in various embodiments an amount of regenerative braking is proportional to an amount of the detected deceleration. In various embodiments, the processor **66** causes the inverter controller **60** to control the inverter **56** to generate an amount of regenerative braking that is proportional to an amount of the detected deceleration.

In various embodiments the instructions are further configured to cause the processor **66** to, responsive to the detected deceleration, during the detected deceleration charge the electrical battery **50** with electrical power generated by electric motors **58** that are rotatable during towing.

In such embodiments, an amount of electrical power supplied to charge the electrical battery is limited to a threshold amount of electrical power—such as an amount of electrical power used by various low voltage electrical loads energizable during towing, such as, for example, various control units, external lighting systems, coolant pumps for cooling the drive units **54** and their components, and the like. For example, in various embodiments the BMU **74** communicates SOC to the ECU **64**. When SOC of the battery **50** has been replenished (such as by the threshold amount of electrical power) to a desired SOC, the processor **66** causes charging of the battery **50** to stop. Thus, various embodiments can help allow avoiding depletion of the battery **50** and still have functionality of the various low voltage electrical loads.

In various embodiments the processor **66** causes the inverter controller **60** to control the inverter **56** to generate zero torque (that is, effect zero mechanical output from the electric motor **58**) when deceleration is no longer detected as described above.

Referring additionally to FIG. **5A**, in various embodiments a method **500** is provided for controlling an inverter of a towed vehicle. The method **500** starts at a block **502**. At a block **504** data indicative of detected deceleration of a tow vehicle is sensed. At a block **506** during detected deceleration, an inverter of a towed vehicle is controlled responsive to the detected deceleration. The method **500** stops at a block **508**.

Referring additionally to FIG. **5B**, in various embodiments at a block **510** the sensed data indicative of detected deceleration of the tow vehicle is received by a controller.

Referring additionally to FIG. **5C**, in various embodiments detecting deceleration of the tow vehicle at the block **510** includes at least one sensing action such as sensing deceleration of the towed vehicle, sensing vehicle speed of the towed vehicle, sensing wheel speed of the towed vehicle, sensing motor speed of the towed vehicle, sensing back emf of an electric motor of the towed vehicle, and/or visually detecting illumination of at least one brake light of the tow vehicle at a block **512**.

Referring additionally to FIG. **5D**, in various embodiments controlling an inverter of a towed vehicle responsive to the detected deceleration at the block **506** includes activating switches in inverters for electric motors that are rotating at a block **514**.

Referring additionally to FIG. **5E**, in various embodiments at a block **516** regenerative braking is applied to the

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towed vehicle. In various embodiments an amount of regenerative braking is proportional to an amount of the detected deceleration.

Referring additionally to FIG. **5F**, in various embodiments at a block **518** an electrical battery of the towed vehicle is charged with electrical power generated by electric motors that are rotating. In various embodiments an amount of electrical power supplied to charge the electrical battery is limited to a threshold amount of electrical power, such as an amount of electrical power used by electrical loads energized during towing.

In some instances, one or more components may be referred to herein as “configured to,” “configured by,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that such terms (for example “configured to”) generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

While particular aspects of the present subject matter described herein have been shown and described, it will be apparent to those skilled in the art that, based upon the teachings herein, changes and modifications may be made without departing from the subject matter described herein and its broader aspects and, therefore, the appended claims are to encompass within their scope all such changes and modifications as are within the true spirit and scope of the subject matter described herein. It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (for example, bodies of the appended claims) are generally intended as “open” terms (for example, the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (for example, “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (for example, the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (for example, “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically

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a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

The foregoing detailed description has set forth various embodiments of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, or examples can be implemented, individually and/or collectively, by a wide range of hardware.

With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flows are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

While the disclosed subject matter has been described in terms of illustrative embodiments, it will be understood by those skilled in the art that various modifications can be made thereto without departing from the scope of the claimed subject matter as set forth in the claims.

What is claimed is:

1. A system comprising: a controller including:
a processor; and
computer-readable media configured to store computer-executable instructions configured to cause the processor to:
receive, from a sensor of a towed vehicle, sensed data indicative of deceleration of a tow vehicle;
detect, using the sensed data, a change in distance between the tow vehicle and the towed vehicle differing from a predetermined distance; and
during detected deceleration and responsive to the change in distance differing from the predetermined distance, control an inverter of the towed vehicle disconnected from the tow vehicle to cause braking of the towed vehicle to be applied to bring the towed vehicle to a stop.
2. The system of claim 1, wherein the sensed data indicative of the deceleration of the tow vehicle includes at least one datum chosen from sensed deceleration of the towed vehicle, sensed vehicle speed of the towed vehicle, sensed wheel speed of the towed vehicle, sensed motor speed of the towed vehicle, sensed back emf of an electric motor of the towed vehicle, and visually detected illumination of at least one brake light of the tow vehicle.
3. The system of claim 1, wherein the instructions are further configured to cause the processor to, responsive to the change in distance between the tow vehicle and the towed vehicle determined from the sensed data of the sensor within the towed vehicle and during the detected deceleration, control activation of switches in for the inventor to control an electric motor that is rotatable during towing.

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4. The system of claim 3, wherein the instructions are further configured to cause the processor to, responsive to the detected deceleration, apply regenerative braking to the towed vehicle during the detected deceleration and synchronize a signal light of the towed vehicle with a signal light of the tow vehicle.

5. The system of claim 3, wherein the instructions are further configured to cause the processor to, responsive to the detected deceleration, during the detected deceleration, charge an electrical battery of the towed vehicle with electrical power generated by electric motors that are rotatable during towing and synchronize braking of the towed vehicle with braking of the tow vehicle.

6. A vehicle comprising:

a plurality of rotatable wheels; at least one drive unit including:

at least one inverter; and

at least one electric motor electrically couplable to the at least one inverter and rotatably couplable to at least one of the plurality of rotatable wheels;

at least one sensor of a towed vehicle; and

a controller including:

a processor; and

computer-readable media configured to store computer-executable instructions configured to cause the processor to:

receive from the at least one sensor of the towed vehicle sensed data indicative of deceleration of a tow vehicle;

detect, using the sensed data, a change in distance between the tow vehicle and the towed vehicle differing from a predetermined distance; and

during detected deceleration and responsive to the change in distance differing from the predetermined distance, control the at least one inverter of the towed vehicle disconnected from the tow vehicle to cause braking of the towed vehicle to be applied to bring the towed vehicle to a stop.

7. The vehicle of claim 6, wherein the at least one sensor includes at least one sensor chosen from an accelerometer, a vehicle speed sensor, a wheel speed sensor, a motor speed sensor, a back emf sensor, and a camera.

8. The vehicle of claim 6, wherein:

the at least one inverter includes a plurality of switches; and

the instructions are further configured to cause the processor to, responsive to the change in distance between the tow vehicle and the towed vehicle determined from the sensed data of the sensor within the towed vehicle and during the detected deceleration, control activation of the plurality of switches for the inverter to control the electric motors that is rotatable during towing.

9. The vehicle of claim 8, wherein the instructions are further configured to cause the processor to, responsive to the detected deceleration, apply regenerative braking during the detected deceleration and synchronize a signal light of the towed vehicle and a signal light of the tow vehicle.

10. The vehicle of claim 9, wherein an amount of regenerative braking is proportional to an amount of the detected deceleration.

11. The vehicle of claim 8, wherein:

the vehicle further comprises a high-voltage, direct current (DC) electrical battery;

the at least one inverter is electrically couplable to the battery; and

the instructions are further configured to cause the processor to, responsive to the detected deceleration, dur-

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ing the detected deceleration, charge the electrical battery with electrical power generated by electric motors that are rotatable during towing.

12. The vehicle of claim **11**, wherein an amount of electrical power supplied to charge the electrical battery is limited to a threshold amount of electrical power.

13. A method comprising:

sensing, by a sensor of a towed vehicle, data indicative of deceleration of a tow vehicle;

detecting, using the data, a change in distance between the tow vehicle and the towed vehicle differing from a predetermined distance; and

during detected deceleration and responsive to the change in distance differing from the predetermined distance, controlling an inverter of a towed vehicle disconnected from the tow vehicle to cause braking of the towed vehicle to be applied to bring the towed vehicle to a stop.

14. The method of claim **13**, further comprising receiving, by a controller, the sensed data indicative of deceleration of the tow vehicle.

15. The method of claim **14**, wherein sensing data indicative of deceleration of the tow vehicle includes at least one

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sensing action chosen from sensing deceleration of the towed vehicle, sensing vehicle speed of the towed vehicle, sensing wheel speed of the towed vehicle, sensing motor speed of the towed vehicle, sensing back emf of an electric motor of the towed vehicle, and visually detecting illumination of at least one brake light of the tow vehicle.

16. The method of claim **13**, wherein controlling an inverter of the towed vehicle responsive to the deceleration includes activating switches for the inverter to control an electric motor that is rotating.

17. The method of claim **16**, further comprising applying regenerative braking to the towed vehicle.

18. The method of claim **17**, wherein an amount of regenerative braking is proportional to an amount of the detected deceleration.

19. The method of claim **16**, further comprising charging an electrical battery of the towed vehicle with electrical power generated by electric motors that are rotating.

20. The method of claim **19**, wherein an amount of electrical power supplied to charge the electrical battery is limited to a threshold amount of electrical power.

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